

- 1- A rocket propellant is stored in a thermally insulated tank at 207 kPa and 85 K. From here, it is supplied to a turbopump. The liquid propellant has a specific gravity of 1.5 and a vapor pressure of 101.3 kPa at 85 K. Given that the propellant enters the turbopump with a speed of 8 m/s, check if cavitation is likely to be a problem. The feed lines are well insulated, and the pump inlet is 1 m below the level of the propellant in the tank. The pipe loss is 0.5 m.
- 2- Question below, which is question 13.3 of the textbook.

Sketched in the figure is a turbopump feed system pumping a total propellant mass flow of 92 kg/s. If the component efficiencies are as listed below, what fraction of the total propellant flow is diverted through the gas generator and turbine?

Fuel pump efficiency	$\eta_{pf} = 0.70$
LOX pump efficiency	$\eta_{PL} = 0.65$
Turbine efficiency	$\eta_{ts} = 0.70$
Turbine discharge pressure	$p_2 = 0.14 \text{ MPa}$
Mechanical efficiency	$\eta_m = 0.90$

Fluid properties are:

$$c_p = 1.07 \text{ kJ/kg} \cdot \text{K} = \text{const. for turbine fluid,}$$

$$\gamma = 1.3 = \text{const. for turbine fluid,}$$

$$\rho_{\text{LOX}} = 1140 \text{ kg/m}^3,$$

$$\rho_{\text{fuel}} = 800 \text{ kg/m}^3.$$

